

Analysis of Data Communication in a Web Application Using Web Development Tools

Aim:

To design and analyze a web application that demonstrates data communication between the client, web server, and database using web development tools such as HTML, CSS, JavaScript, a backend framework, and HTTP/HTTPS protocols.

Objectives:

- To understand the concept of data communication in web applications.
- To study the interaction between the **client, web server, and database**.
- To develop a simple web application using HTML, CSS, JavaScript, and a suitable backend.
- To transmit data between frontend and backend using HTTP requests.
- To analyze how data is received, processed, and displayed on the web page.
- To understand the importance of data formats such as JSON in web communication.
- To test the accuracy and reliability of data transmission.

Problem Statement:

Modern web applications continuously exchange data between the user's browser and the server. When a user submits information through a web page, the data must be transferred securely and accurately to the server, processed, and returned to the user.

The problem is to design a web application that demonstrates this communication process and analyzes how data flows between the frontend, backend, and database. The application should be capable of sending requests, receiving responses, and displaying the processed data to the user.

Proposed System:

The proposed system is a simple web application that allows a user to enter information through a web interface.

A web application can be developed using the following technologies:

- **HTML** – Creates the structure of the web page.
- **CSS** – Provides styling and layout.
- **JavaScript** – Handles user interaction and sends/receives data.
- **Web server/backend** – Processes client requests.
- **Database** – Stores and retrieves application data.
- **HTTP/HTTPS** – Provides communication between client and server.
- **JSON** – Used to exchange structured data between frontend and backend.

The basic communication process is:

User → Web Browser → HTTP Request → Web Server → Database → Web Server → HTTP Response → Web Browser → User

Tools and Technologies Used:

Tool/Technology	Purpose
HTML	Creates webpage structure
CSS	Designs and styles the webpage
JavaScript	Provides interactivity and communication
HTTP/HTTPS	Transfers data between client and server
JSON	Represents data exchanged between client and server
Backend	Processes client requests
Database	Stores application data
Web Browser	Acts as the client
VS Code/IDE	Used for developing the application
Developer Tools	Used to analyze network communication

Result:

The web application successfully demonstrates data communication between the client and server. Data entered by the user is transmitted through an HTTP request, processed by the backend, and returned to the frontend as a response.

The analysis shows that web development tools provide an effective method for implementing real-time data exchange, request-response communication, data processing, and database interaction. Thus, the developed application demonstrates the complete flow of data communication in a web-based system.